SRINIVASAN VIJAYARAGHAVAN

DevOps / Site Reliability Engineer — Cloud Infrastructure, Reliability, Release Automation & AI Developer Tooling

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U.S. Citizen · OCI Holder — authorized to work in the United States and India

SUMMARY

DevOps / Site Reliability Engineer with 5 years of experience on Azure and AWS. At Thomson Reuters, own release, upgrade and disaster-recovery automation for a multitenant Azure platform of 15+ microservices, built on GitHub Actions, Ansible and Datadog. Previously set up build, packaging and release pipelines from scratch for embedded and enterprise products on Linux and Windows. Also maintain repo2graph, an open-source MCP server that helps AI coding agents understand codebases.

EXPERIENCE

DevOps Engineer — Thomson Reuters

3 yrs 3 mos · Bangalore, India

- Built the parallelized Azure Storage migration framework that the platform's large-scale data transfers run on: custom runners, chunked delta detection and parallel AzCopy orchestration, with SAS-authenticated Azure Table Storage driving inventory generation and migration-summary reporting.
- Orchestrated disaster-recovery failover and failback through Azure Service Bus with structured Datadog logging, holding zero data loss across same-region and cross-region Azure migrations.
- Replaced manual multi-page runbooks with a self-service GitHub Actions suite covering instance provisioning, configuration promotion, multi-instance operations and regional failover.
- Eliminated manual upgrade runbooks across 15+ microservices with unattended Ansible orchestration — feature-flag loops, rescue blocks and automatic rollback — and refactored fragmented inventories into a single environment-agnostic model.
- Designed a fail-fast Apache health gate validating four conditions before traffic is routed (service state, HTTP 200, load-balancer convergence, health-check file), with drain timeouts and retry/backoff at 5 attempts on a 2-second delay to stop false restarts.
- Gated releases on T-SQL-verified backups with RESTORE-based rollback and Datadog failure hooks, plus a safe-skip flow that polls active SQL Server tasks before backup verification so deployments never block on a live job.
- Built rollback and recovery pipelines handling database restoration, service resets, checkpoint cleanup and external system synchronisation, including DNS and MX cutovers automated in Python against the Akamai API, and Pingdom check updates.
- Converted the estate's ARM templates to Bicep and automated the conversion path end to end, standardising infrastructure definitions across environments.
- Automated container image promotion in GitHub Actions with real-time Azure DevOps, ACR and AKS log polling, and closed a security gap with two RBAC-gated workflows plus BYOK encryption through Key Vault.
- Centralised variable and secret management (vars.yml / vault.yml) across environments, automated cross-environment config syncing (duplicate-key cleanup, lifecycle updates, ID normalisation), and drove conditional BYOK configuration through Jinja2 templating and Azure App Config.
- Extended HTTP, TCP and JMX monitoring to every microservice with dynamic templates, standardised tagging and dependency-aware restarts, wiring Datadog dashboards and Teams alerting directly into deployment pipelines.
- Stood up on-premises deployment from zero tooling using Jinja2 templates, idempotent SQL Server initialisation and version-gated JVM provisioning for Spring Boot migrations, and built shared-resource synchronisation frameworks for Solr and application servers.
- Led a jumpbox VM SKU and family upgrade through formal ITSM change management, moving the fleet to better-value instance types.

DevOps Engineer — GraniteRiverLabs

1 yr 9 mos · Bangalore, India

- Sole engineer for build and release tooling across Linux, Windows and cloud, serving both embedded and enterprise product teams; promoted at six months.
- Standardised release delivery with Docker build, tag, push and deploy pipelines for frontend and backend services, taking releases from manual steps to one-click.
- Automated WiX installer builds and code signing in GitHub Actions to produce release-ready signed applications, and ran dpkg packaging in CI for Linux builds.
- Authored a custom Windows Wireshark installer for THREAD protocol analysis, delivered through GitLab CI; the work was adopted after my departure and merged into the upstream Wireshark project.

- Delivered deployment pipelines and hardware-in-the-loop test automation for Project MATTER (Connectivity Standards Alliance) and Zigbee smart-device interoperability testing.
- Contributed C#/.NET and C/C++ production code to the company's own product line alongside the release tooling.
- Automated ARM-based machine image builds with tuned configurations, and wrote Bash tooling for system performance monitoring.
- Restructured GitHub repositories to enforce linting, unit tests, code formatting and PR quality gates, raising review consistency across the team.
- Scoped AWS EC2 hosting for .NET applications to least privilege through security group rules and key-pair management.
- Recognised twice: for Zigbee/Matter team support and on-call effort, and for product build and automation ownership.

AI PROJECTS

repo2graph — Creator & Maintainer

Open source (MIT) · PyPI · GitHub Marketplace · github.com/Srinivasan-78/repo2graph

- Open-source code-graph engine and MCP server that gives AI coding agents (Claude Code, Cursor) cited, token-bounded answers about any codebase — no LLM or API key needed.
- Tree-sitter parsing across 17 languages, BM25 + graph retrieval, and a GitHub Action for PR blast-radius analysis; shipped on PyPI, npm and Docker.

Offline RAG Knowledge Assistant — Homelab

- Local chatbot over offline Wikipedia and Arch Wiki using FastAPI, Ollama (Llama 3.2) and Streamlit.

CORE COMPETENCIES

AI & LLM Engineering: LLM API integration, Model Context Protocol (MCP) servers, RAG (BM25, embeddings), agentic coding workflows, prompt engineering, local LLM hosting

AI Tools & Models: Claude Code, Claude API, Cursor, GitHub Copilot, ChatGPT / OpenAI API, Google Gemini, Ollama, Llama, DeepSeek, Groq, OpenRouter, Hugging Face

CI/CD & IaC: GitHub Actions, GitLab CI, Azure DevOps (incl. REST API), Terraform, Ansible, Jinja2, Bicep, ARM, Docker, Kubernetes (AKS), WiX Toolset, dpkg

Cloud: Azure (Storage, AzCopy, Service Bus, App Config, Key Vault, ACR, AKS, Load Balancer, CLI), AWS (EC2, Security Groups)

Reliability & Operations: DR failover/failback, SQL Server HA mirroring, backup/restore verification, health-gated restarts, zero-downtime traffic shifts, ITSM change management

Monitoring: Datadog, Pingdom, HTTP/TCP/JMX checks, deployment telemetry, Teams alerting

Platforms: Windows Server, RHEL, Ubuntu, Apache, Tomcat, SQL Server (T-SQL), Solr, WinRM, Akamai

Languages: Python, Bash/Shell, PowerShell, Batch, SQL, YAML, C/C++, C#

Security: RBAC in CI/CD, secret management, BYOK, S/MIME commit signing, vulnerability scanning